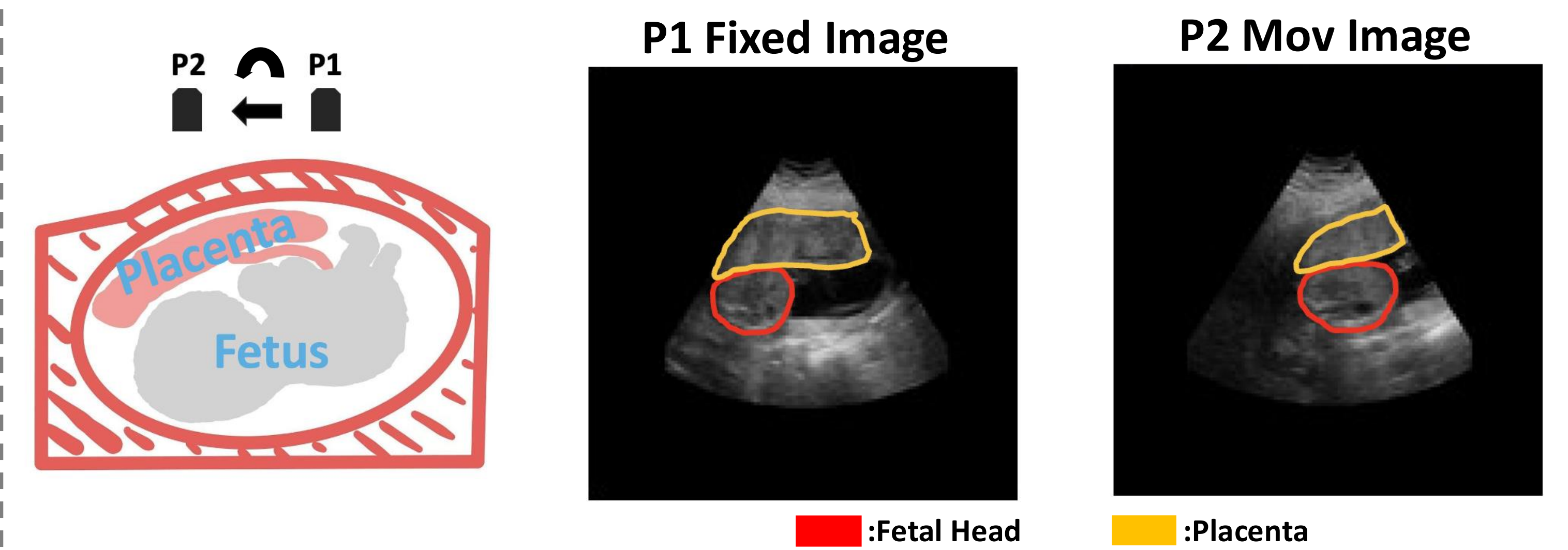


From Geometry to Intensity: A Coarse-to-Fine Pipeline for Unsupervised 3D Ultrasound Stitching

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Introduction – 3D Ultrasound (3DUS) Stitching

3DUS stitching expands the field-of-view (FOV) by aligning images collected from different probe positions.

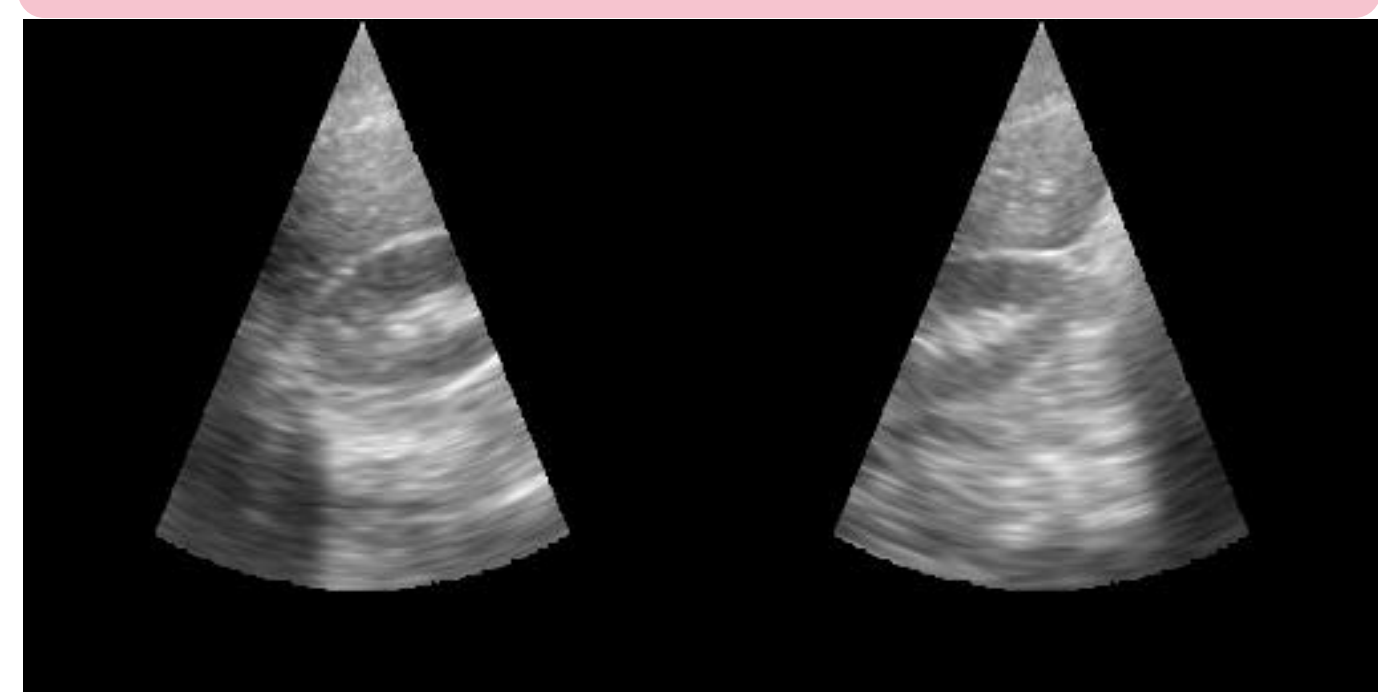


- Inherent trade-off between spatial/temporal resolution and FOV
- Target anatomical structure may be larger than FOV
- **US stitching** is needed

Challenges

Challenge1: Extensive Motion Between Stitching Pairs

Large Translation[1]

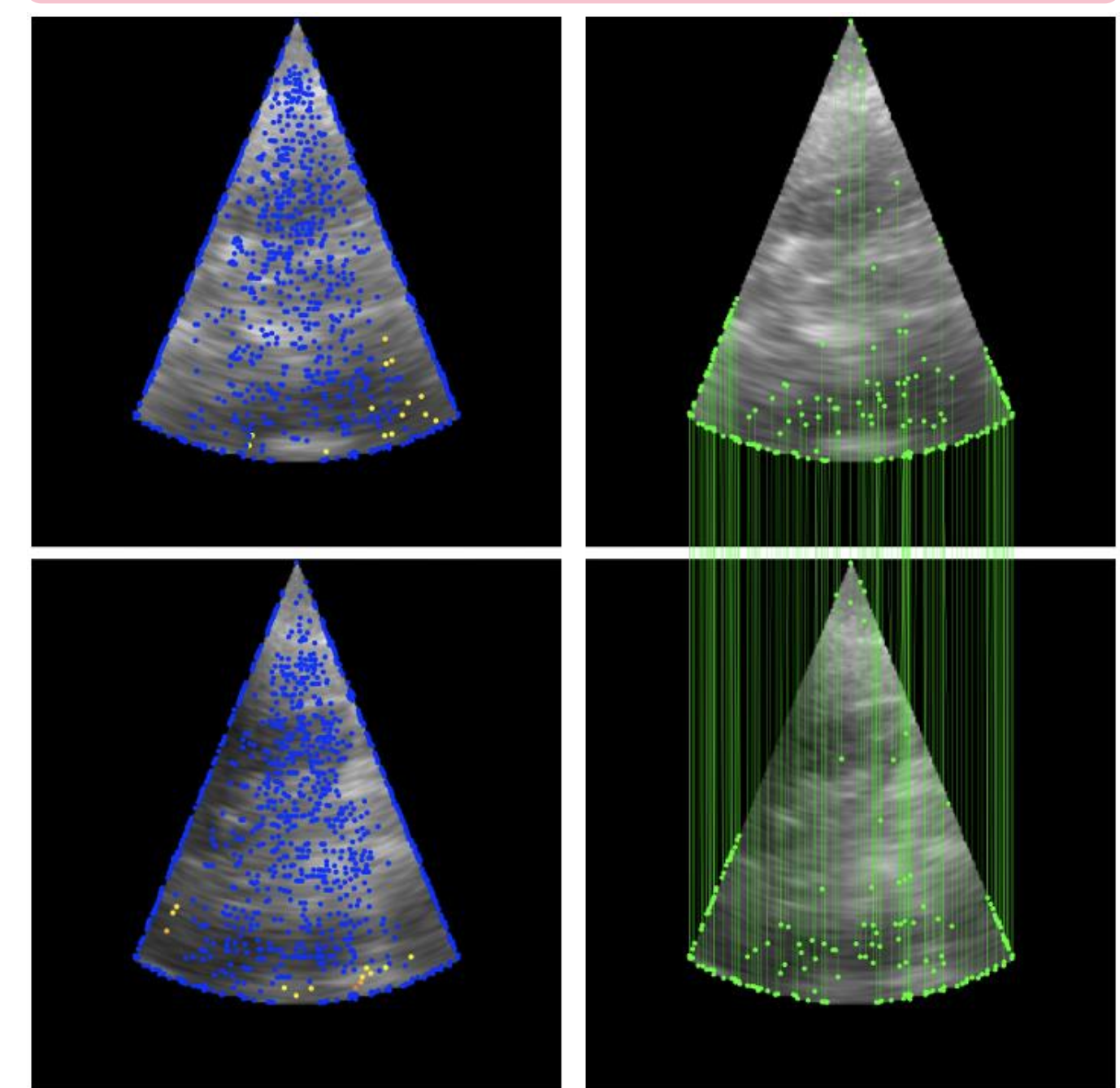


Large Rotation[2]



Challenge2: Sector-Shaped FOV Impact

Edge Guides the Stitching[1]



Stuck into Local Minima during Optimization[2]

$$NCC(A, \text{Reg}(B)) = \frac{\sum_i (A_i - \bar{A})(\text{Reg}(B)_i - \text{Reg}(\bar{B}))}{\sqrt{\sum_i (A_i - \bar{A})^2 \sum_i (\text{Reg}(B)_i - \text{Reg}(\bar{B}))^2}} \quad (1)$$

$$MSE(A, \text{Reg}(B)) = \frac{1}{N} \sum_{i=1}^N (A_i - \text{Reg}(B)_i)^2 \quad (2)$$

Fix Image Mov Image Results



Challenge3: Data Limitation

Limited Open-Source Datasets

Limited Scope of Tasks

Human-Annotation

In MedSAM2
US : Others
16 : 220

Mainly focusing on segmentation task

Time- and Labor-consuming

Datasets

For registration testing, 40 registration pairs from 20 subjects are used

- Max rotation: [30°, 117°], max trans: [25mm, 83mm]
- Hard Case: 20 pairs, rotation > 80° or translation > 60mm
- Ground Truth: 2 raters manual annotation + 3 raters validation

Methods

Training-Free Coarse-to-Fine 3DUS Stitching Pipeline

Unsupervised Point Cloud Extraction

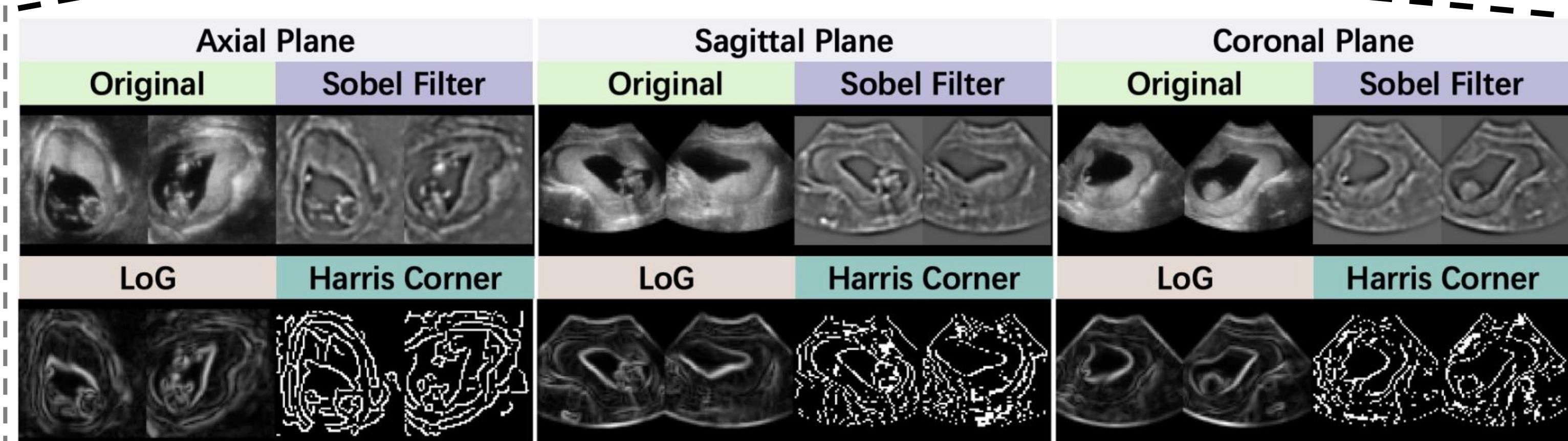
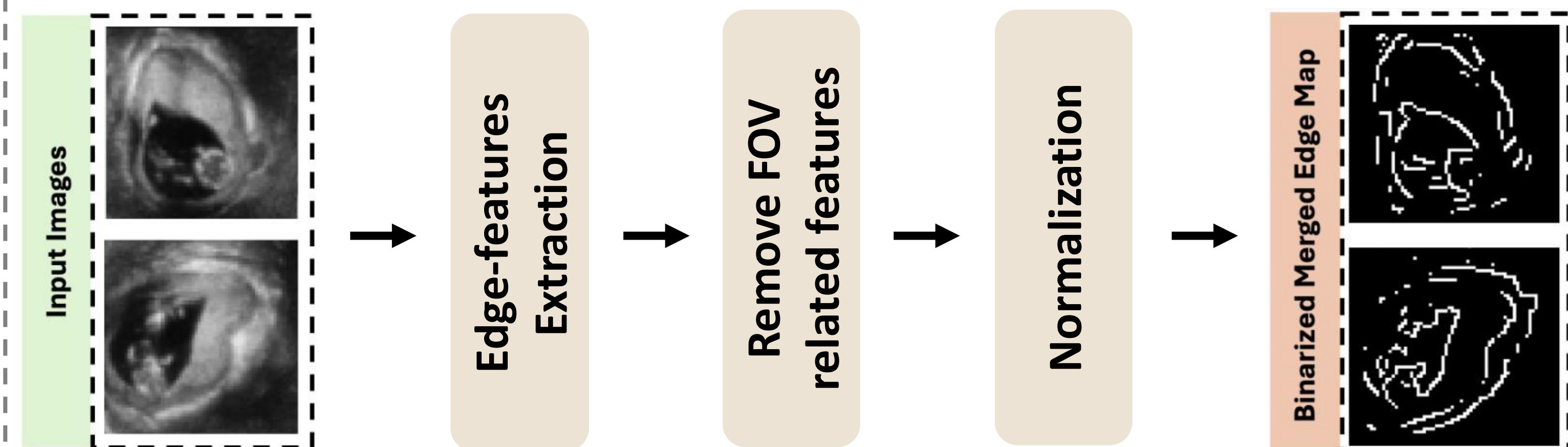


Point cloud (PCD)-based coarse registration



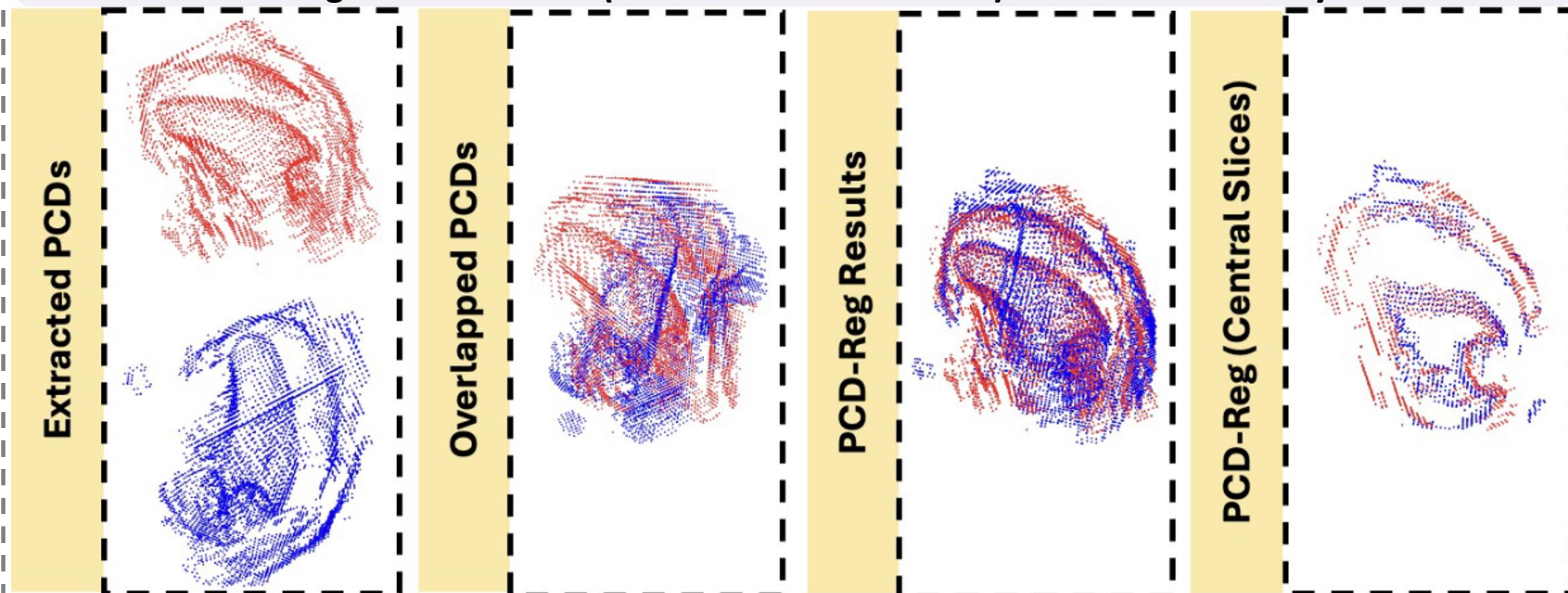
Intensity-based fine registration

Step1: Unsupervised Point Cloud Extraction



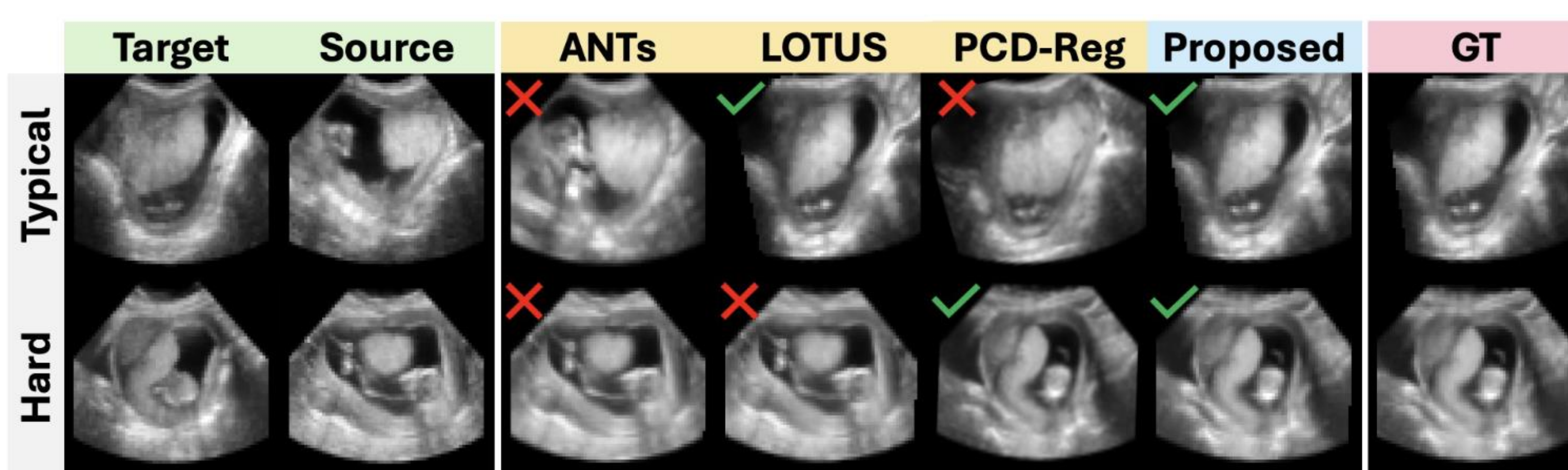
Step2: PCD-Based Coarse Registration (before ANTs[3] refinement)

From left to right: the extracted PCDs, overlapped PCDs before registration, and the visualization of registration results (before ANTs refinement) in all slices and only central slices



Results

Qualitative Results: proposed method outperforms baselines in both subjects



Quantitative Results for 3DUS registration

Model	NCC ↑	SSIM ↑	MSE ↓
ANTs	0.826±0.106	0.496±0.219	0.0174±0.0096
LOTUS	0.881±0.132	0.608±0.251	0.0120±0.0110
PCD-Reg	0.847±0.140	0.539±0.201	0.0144±0.0093
Proposed	0.889±0.125*	0.654±0.262*	0.0105±0.0098*

Conclusion

- The proposed method outperforms conventional and learning-based baselines without supervised training.

Acknowledgements

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Reference

- [1] Yao X, et al. ISBI 2025
- [2] Yao X, et al. MIDL 2025
- [3] Avants BB, et al. MIA 2028

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Follow-Up Work

Curious about fusing vision foundation model-derived semantic features with PCD-based geometric features?

